Transformer Block: Single Unified Diagram ($d_{\text{model}} = 3$, 2 heads)

Encoder-style, post-LN, pedagogical example. One continuous flow with all key details embedded directly in the pipeline.

Input preparation

Token embeddings

$\mathbf{E} \in \mathbb{R}^{3\times3}$

Token 1 $\mathbf{x}_1 = \begin{bmatrix}0.1\\0.5\\1.2\end{bmatrix}$

Token 2 $\mathbf{x}_2 = \begin{bmatrix}1.0\\0.3\\0.7\end{bmatrix}$

Token 3 $\mathbf{x}_3 = \begin{bmatrix}0.4\\1.1\\0.9\end{bmatrix}$

Sequence length = 3
Each token is a 3D vector.

Positional encodings

$\mathbf{P} \in \mathbb{R}^{3\times3}$

$p_1 = \begin{bmatrix}0.2\\0.1\\0.0\end{bmatrix}$

$p_2 = \begin{bmatrix}0.0\\0.2\\0.1\end{bmatrix}$

$p_3 = \begin{bmatrix}0.1\\0.0\\0.2\end{bmatrix}$

$+$

Position-aware input

$\mathbf{X} = \mathbf{E} + \mathbf{P} \in \mathbb{R}^{3\times3}$

Token 1 $\mathbf{X}_1 = \begin{bmatrix}0.3\\0.6\\1.2\end{bmatrix}$

Token 2 $\mathbf{X}_2 = \begin{bmatrix}1.0\\0.5\\0.8\end{bmatrix}$

Token 3 $\mathbf{X}_3 = \begin{bmatrix}0.5\\1.1\\1.1\end{bmatrix}$

X is a 3 × 3 sequence matrix.

Transformer block

Multi-Head Self-Attention (2 heads, encoder-style, unmasked)

Bidirectional self-attention: each token attends to all tokens (no causal mask).

Illustrative numerical scores/weights; learned Q/K/V projection weights are omitted.

Input X

$\mathbf{X} \in \mathbb{R}^{3\times3}$

0.3	0.6	1.2
1.0	0.5	0.8
0.5	1.1	1.1

rows = tokens
cols = features

This 3 × 3 tensor
is projected into Q, K, V
inside every head.

Head 1 (independently projects 3D → 3D)

$\mathbf{Q}^{(1)} = \mathbf{X}\mathbf{W}_Q^{(1)}$ (3 × 3)

$\mathbf{K}^{(1)} = \mathbf{X}\mathbf{W}_K^{(1)}$ (3 × 3)

$\mathbf{V}^{(1)} = \mathbf{X}\mathbf{W}_V^{(1)}$ (3 × 3)

Scores
$\mathbf{S}_1 = \mathbf{Q}^{(1)}\mathbf{K}^{(1)\mathsf{T}}$

$$\begin{bmatrix}1.2 & 0.1 & -0.3\\0.0 & 0.8 & 0.2\\-0.1 & 0.4 & 1.1\end{bmatrix}$$

Scale
$\hat{\mathbf{S}}_1 = \mathbf{S}_1/\sqrt{3}$
(3 × 3)

Row-wise softmax
$\mathbf{A}_1 = \mathrm{softmax}(\hat{\mathbf{S}}_1)$
(3 × 3)

$$\begin{bmatrix}0.513 & 0.272 & 0.216\\0.270 & 0.428 & 0.303\\0.231 & 0.308 & 0.461\end{bmatrix}$$

rows sum to 1

Weighted sum
$\mathbf{H}_1 = \mathbf{A}_1\mathbf{V}^{(1)}$
(3 × 3)

$\mathbf{H}_1$ **(3 × 3)**

Head 2 (independently projects 3D → 3D)

$\mathbf{Q}^{(2)} = \mathbf{X}\mathbf{W}_Q^{(2)}$ (3 × 3)

$\mathbf{K}^{(2)} = \mathbf{X}\mathbf{W}_K^{(2)}$ (3 × 3)

$\mathbf{V}^{(2)} = \mathbf{X}\mathbf{W}_V^{(2)}$ (3 × 3)

Scores
$\mathbf{S}_2 = \mathbf{Q}^{(2)}\mathbf{K}^{(2)\mathsf{T}}$

$$\begin{bmatrix}0.3 & -0.2 & 0.8\\-0.1 & 1.2 & 0.0\\0.5 & 0.1 & -0.4\end{bmatrix}$$

Scale
$\hat{\mathbf{S}}_2 = \mathbf{S}_2/\sqrt{3}$
(3 × 3)

Row-wise softmax
$\mathbf{A}_2 = \mathrm{softmax}(\hat{\mathbf{S}}_2)$
(3 × 3)

$$\begin{bmatrix}0.324 & 0.243 & 0.433\\0.239 & 0.507 & 0.254\\0.419 & 0.332 & 0.249\end{bmatrix}$$

rows sum to 1

Weighted sum
$\mathbf{H}_2 = \mathbf{A}_2\mathbf{V}^{(2)}$
(3 × 3)

$\mathbf{H}_2$ **(3 × 3)**

Residual 1: X

Concatenate feature columns

$\mathbf{H}_1$ $\mathbf{H}_2$

3 × 3 + 3 × 3
→ 3 × 6

$\mathbf{H}_{\text{cat}} = [\mathbf{H}_1 \parallel \mathbf{H}_2]$

Output projection

$\mathbf{W}_O \in \mathbb{R}^{6\times3}$

learned feature mix
6 → 3

Multi-head attention output

$\mathbf{Z}_{\text{attn}} = \mathbf{H}_{\text{cat}}\mathbf{W}_O$

context-mixed
3 × 3 output

$\mathbf{Z}_{\text{attn}} \in \mathbb{R}^{3\times3}$

$+$

LayerNorm 1
(3 × 3)

Position-wise Feed-Forward Network

Same MLP applied independently to each token row.

Linear 1
3 → 12 features
for each token

ReLU
max(0, x)

-0.8	0.3	-0.1
0	0.3	0

example subset
elementwise

Linear 2
12 → 3 features
for each token

3 × 3 → 3 × 12 → 3 × 12 → 3 × 3

Same weights for Token 1/2/3; no token-to-token mixing in the FFN.

Residual 2: LN1 output

$+$

LayerNorm 2
(3 × 3)

Output sequence

$\mathbf{z}_i \in \mathbb{R}^3$

(illustrative output vectors)

Token 1 $\mathbf{z}_1 = \begin{bmatrix}0.2\\0.8\\1.0\end{bmatrix}$

Token 2 $\mathbf{z}_2 = \begin{bmatrix}1.1\\0.1\\0.5\end{bmatrix}$

Token 3 $\mathbf{z}_3 = \begin{bmatrix}0.6\\1.0\\0.8\end{bmatrix}$

$\mathbf{Z} \in \mathbb{R}^{3\times3}$

(3 tokens × 3 dim)